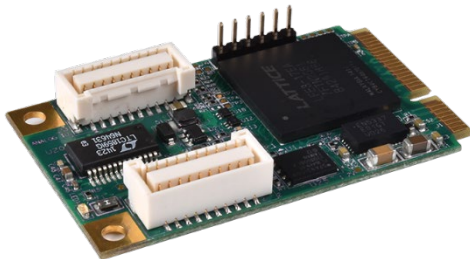


IO183

Simultaneous Sampling 16-bit Analog I/O Module



Included in the Delivery

- I/O module installed in the real-time target machine
- Cable
- Terminal board
- Simulink driver blocks
- Test models
- Comprehensive documentation

Capabilities

The IO183 I/O module is a fast, simultaneous-sampling, 16-bit analog input and output module with Simulink® driver blocks.

The module features eight single-ended (or four differential) analog inputs, four single-ended analog outputs and 14 digital I/O lines.

The IO183 I/O module can be connected by trigger lines to additional I/O modules to synchronously sample and convert analog input or output channels across the different modules.

This I/O module is ideal for closed-loop controls and hardware-in-the-loop simulations using MATLAB® and Simulink.

Common Applications

- Rapid control prototyping
- Hardware-in-the-loop (HIL) simulation

Supported Target Machines

- Baseline

Key Features

Analog Inputs	Analog Outputs	Digital I/O
8 single-ended or 4 differential inputs	4 single-ended outputs	14 digital I/O lines individually configurable as input or output
16-bit resolution	16-bit resolution	—
Software selectable ± 10 V, ± 5 V, 0-10 V, 0-5 V input range (one setting for all channels)	Software selectable 0-2.5 V, 0-5 V output range (one setting for all channels), single channel and simultaneous update modes	Software selectable pull-up/-down resistors
Simulink Real-Time™ analog input driver blocks	Simulink Real-Time analog output driver blocks	Simulink Real-Time digital input and output driver blocks

Technical Specifications

Physical	
Form factor	mPCIe
Power requirements	0.462 W @ 3.3 V DC
Bus	PCI Express x1
Connectors	Digital I/O: 17-pin male M12 connector Analog I/O: 17-pin female M12 connector
Environmental	
Operating temperature	-40 °C to +85 °C
Relative humidity	5 to 95 %, non-condensing
Analog Input	
Number of inputs	8x 16-bit single ended or 4x 16-bit differential
A/D resolution	16 bits
Sample rate	100 kHz maximum aggregate
Voltage ranges	Software selectable ± 10 V, ± 5 V, 0-10 V, 0-5 V (one setting for all channels)
Input over voltage protection	Up to ± 25 V
Input impedance	31K Ω typical (for bipolar input ranges), 42K Ω typical (for unipolar input ranges)
Analog Output	
Number of outputs	4 single-ended
D/A resolution	16 bits
Voltage ranges	Software selectable 0-2.5 V, 0-5 V output range (one setting for all channels)
Output current	Up to 5 mA per channel (1K Ω minimum load)
Output error	± 1.5 mV (± 0.03 %) max zero scale error (no offset adjustment)
Digital I/O	
Configuration	14 lines buffered, individual direction control and protection
Output voltage	+3.3 V

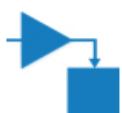
Order and Contact Information

Item ID	Product Name	Components
201838	IO183	1x IO183 I/O module
		2x 17-Pin Male M12 to 17-Pin Female M12 I/O Cable (3 ft / 0.91 m)
		1x 2-way 17-Pin M12 Terminal Board
		Driver block library for Simulink Real-Time
		Comprehensive documentation and Simulink example models
		Installation into the real-time target machine
Optional Accessories		
20183N	IO183-BNC-Box	Break-out BNC box for IO183 I/O module providing a total of 8 BNC connectors to access 8 single-ended or 4 differential analog inputs, 4 BNC connectors for 4 single-ended analog outputs, and 16 BNC connectors to access 16 digital TTL I/O lines

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Benefits of Speedgoat Solutions



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